

Analysis

Country-wise infection count: The data set contains the state-wise infection count of some countries. To have a country-wise count of infection on each date, it is necessary to make a group of all the states of a country. This can be done in different ways.

Here I have used the *aggregate()* function to perform this group-wise summation, as shown below:

```
dfCountryGroupSum <- aggregate(df[,c(5:
ncol(df))],by=list(Category=df$Country.
Region),FUN=sum)
>head(dfCountryGroupSum[dfCountryGroupS
um$Category=="China" ,c(1:5)])
#   Category X1.22.2020 X1.23.2020
X1.24.2020 X1.25.2020
#28   China      548      643
920      1406
>dfCountryGroupSum[dfCountryGroupSum$Ca
teory=="China" | dfCountryGroupSum$Cat
egory=="India",c(1:10)]
   Category X1.22.20 X1.23.20 X1.24.20
X1.25.20 X1.26.20 X1.27.20 X1.28.20
X1.29.20 X1.30.20
37   China      548      643      920
1406      2075      2877      5509
6087      8141
80   India        0        0        0
0        0        0        0        0
1
> head(dfCountryGroupSum[dfC
ountryGroupSum$Category %in%
c("China","India"),c(1:5)])
#28   China      548      643
920      1406
#63   India        0        0
0        0
```

The aggregated result is stored in a dataframe variable *dfCountryGroupSum*, and displayed with dataframe indexing with the corresponding variables and logical filtering.

Aggregate of total infection of each country: To get the aggregate of infection in each country, it is necessary to add all the date-wise infection counts. This is implemented here

with *apply()*. The *sum()* function is mapped to each row from the second column to the last column of dataframe *dfCountryGroupSum*.

```
>dfCountryGroupSum$Total <- apply(dfCou
ntryGroupSum[,c(2:ncol(dfCountryGroupSu
m))],1,sum,na.omit=T)
> head(dfCountryGroupSum[with(dfCountry
GroupSum, Category %in% c("China","Indi
a")),c(1,ncol(dfCountryGroupSum))])
Category   Total
1   China 15452689
2   India 69692867
>View(dfCountryGroupSum)
```

Graphical representation

To have a better view of the analysis, it is always desirable to depict the analytical outcomes in graphical forms.

Using the code below, I can display the exploratory data analysis results in histograms so that a visual

comparative study can be done for the data.

```
>names(dfCountryGroupSum)
[1]<-c("Country")
>TotConfirmedCase <- dfCountryGroupSum[,
c(1,ncol(dfCountryGroupSum))]
>View(TotConfirmedCase)
>p<-ggplot(TotConfirmedCase,
aes(x=Country,y=log(TotCase))) +
  geom_bar(stat="identity")
> p
```

Trend analysis

Trend analysis is important for future projections. As the acquired data contains sufficient reliable day-wise information of confirmed infection cases, we can perform trend analysis using a time series.

For simplicity, I shall demonstrate the time series analysis on some selected records only.

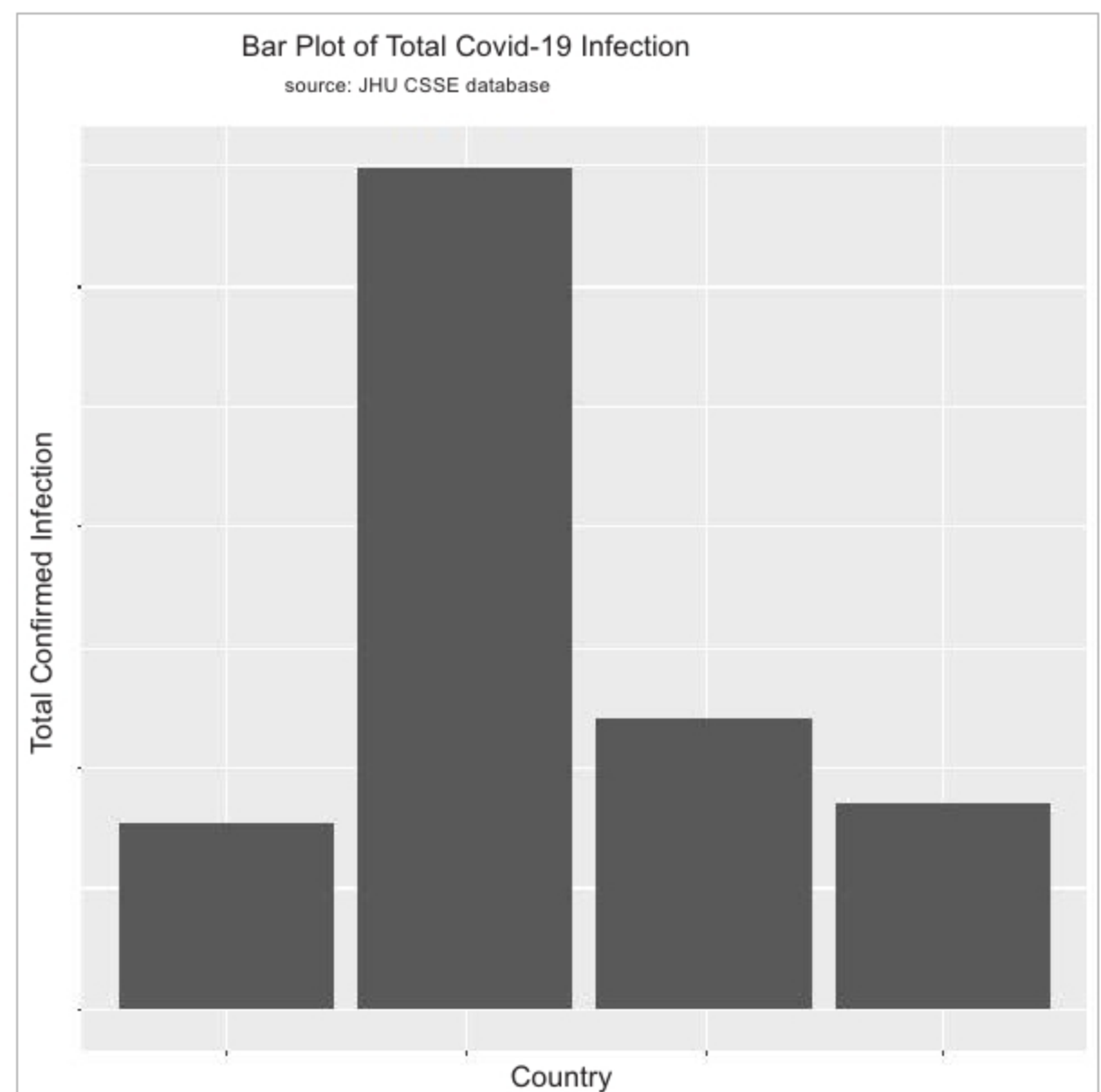


Figure 1: Bar plot to show country-wise total infection count